**DBMS LAB 6**

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## Experiment 5: Implementation of Various Join Operations

**Aim:** Understanding and applying INNER JOIN, LEFT OUTER JOIN, RIGHT OUTER JOIN, FULL OUTER JOIN, SELF JOIN, CROSS JOIN, and Multi-table Joins on relational schemas.

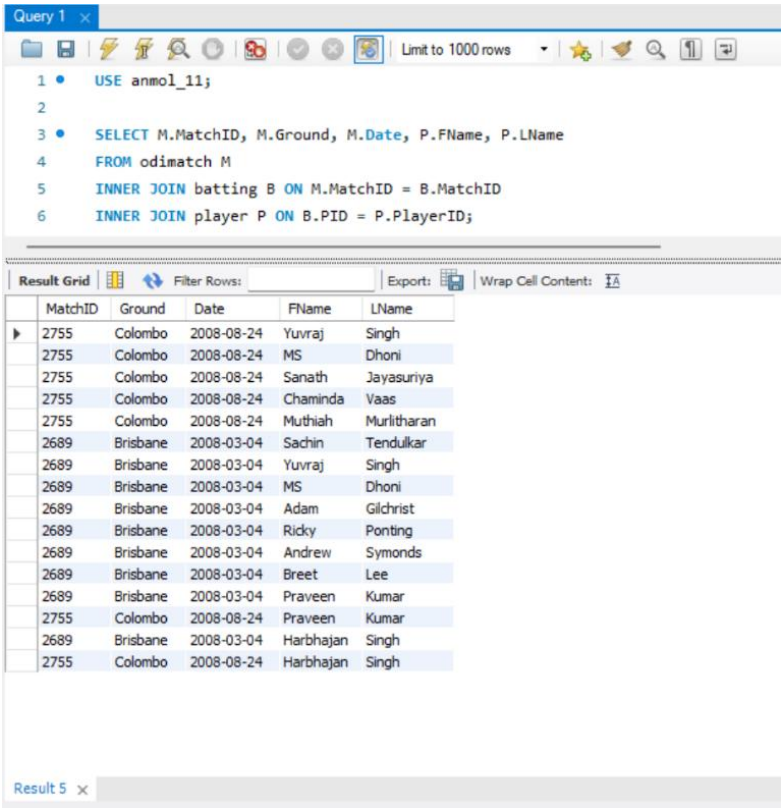
**1 [Inner Join]:** Write an SQL query using INNER JOIN to display the MatchID, Ground, Date, and the Player's full name (FName, LName) for every player who batted in each match.

SELECT M.MatchID, M.Ground, M.Date, P.FName, P.LName

FROM odimatch M

INNER JOIN batting B ON M.MatchID = B.MatchID

INNER JOIN player P ON B.PID = P.PlayerID;




**2 [Inner Join with Filter]:** Display Player full name, Country, NRuns, and Six for all batting performances where the player scored at least one six (Six > 0).

SELECT P.FName, P.LName, P.Country, B.NRuns, B.Six

FROM player P

INNER JOIN batting B ON P.PlayerID = B.PID

WHERE B.Six > 0;

```
USE anmol_11;

SELECT P.FName, P.LName, P.Country, B.NRuns, B.Six
FROM player P
INNER JOIN batting B ON P.PlayerID = B.PID
WHERE B.Six > 0;
```

| FName | LName | Country | NRuns | Six |
|---|---|---|---|---|
| Sanath | Jayasuriya | Sri Lanka | 60 | 2 |
| Yuvraj | Singh | India | 38 | 2 |
| MS | Dhoni | India | 36 | 1 |
| Andrew | Symonds | Australia | 42 | 1 |

**3 [Left Outer Join]:** Write a query using LEFT OUTER JOIN to list all players (PlayerID, FName, LName) along with their batting scores (MatchID, NRuns). Ensure players who have never batted in any match are also displayed with NULL values.

SELECT P.PlayerID, P.FName, P.LName, B.MatchID, B.NRuns

FROM player P

LEFT JOIN batting B ON P.PlayerID = B.PID;

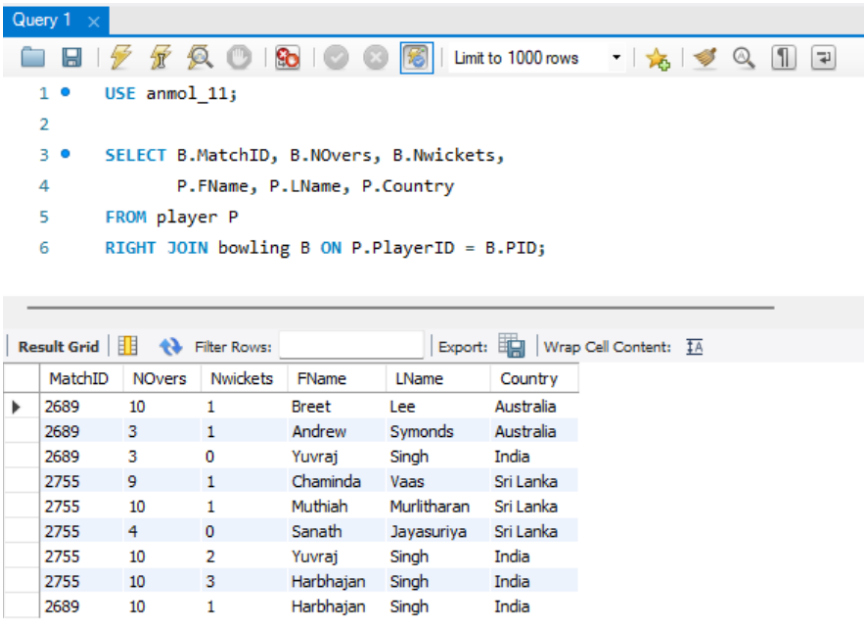
```
USE anmol_11;

SELECT P.PlayerID, P.FName, P.LName, B.MatchID, B.NRuns
FROM player P
LEFT JOIN batting B ON P.PlayerID = B.PID;
```

| PlayerID | FName | LName | MatchID | NRuns |
|---|---|---|---|---|
| 21001 | Virender | Sehwag | NULL | NULL |
| 23001 | Yuvraj | Singh | 2755 | 0 |
| 23001 | Yuvraj | Singh | 2689 | 38 |
| 24001 | Andrew | Symonds | 2689 | 42 |
| 25001 | MS | Dhoni | 2755 | 71 |
| 25001 | MS | Dhoni | 2689 | 36 |
| 27001 | Praveen | Kumar | 2689 | 7 |
| 27001 | Praveen | Kumar | 2755 | 2 |
| 27002 | Ishant | Sharma | NULL | NULL |
| 89001 | Sachin | Tendulkar | 2689 | 91 |
| 90001 | Brain | Lara | NULL | NULL |
| 90002 | Anil | Kumble | NULL | NULL |
| 91001 | Sanath | Jayasuriya | 2755 | 60 |
| 92001 | Shane | Warne | NULL | NULL |
| 92002 | Muthiah | Murlitharan | 2755 | 1 |
| 92003 | Inzamam | UI-Huq | NULL | NULL |
| 93002 | Heath | Streak | NULL | NULL |
| 93003 | Gary | Kirsten | NULL | NULL |
| 94002 | Chaminda | Vaas | 2755 | 17 |
| 94004 | Stephen | Fleming | NULL | NULL |
| 95001 | Ricky | Ponting | 2689 | 1 |
| 95002 | Shaun | Pollock | NULL | NULL |

**4 [Right Outer Join]:** Retrieve all bowling performance entries (MatchID, NOvers, Nwickets) along with player details (FName, LName, Country) using a RIGHT OUTER JOIN, showing all bowling entries even if player profiles are missing.

```
SELECT B.MatchID, B.NOvers, B.Nwickets,
    P.FName, P.LName, P.Country
FROM player P
RIGHT JOIN bowling B ON P.PlayerID = B.PID;
```




| MatchID | NOvers | Nwickets | FName | LName | Country |
|---|---|---|---|---|---|
| 2689 | 10 | 1 | Breet | Lee | Australia |
| 2689 | 3 | 1 | Andrew | Symonds | Australia |
| 2689 | 3 | 0 | Yuvraj | Singh | India |
| 2755 | 9 | 1 | Chaminda | Vaas | Sri Lanka |
| 2755 | 10 | 1 | Muthiah | Murlitharan | Sri Lanka |
| 2755 | 4 | 0 | Sanath | Jayasuriya | Sri Lanka |
| 2755 | 10 | 2 | Yuvraj | Singh | India |
| 2755 | 10 | 3 | Harbhajan | Singh | India |
| 2689 | 10 | 1 | Harbhajan | Singh | India |

**5 [Full Outer Join / Union]:** Write an SQL query to perform a FULL OUTER JOIN (or simulate using LEFT JOIN UNION RIGHT JOIN) between Batting and Bowling tables on MatchID and PID to view all combined match participation records.

```
SELECT B.MatchID, B.PID, B.NRuns, W.NOvers, W.Nwickets
FROM batting B
LEFT JOIN bowling W
ON B.MatchID = W.MatchID AND B.PID = W.PID
UNION
SELECT B.MatchID, B.PID, B.NRuns, W.NOvers, W.Nwickets
FROM batting B
RIGHT JOIN bowling W
ON B.MatchID = W.MatchID AND B.PID = W.PID;
```

Query 1

Limit to 1000 rows

```
1  USE anmol_11;
2
3  SELECT B.MatchID, B.PID, B.NRuns, W.NOvers, W.Nwickets
4  FROM batting B
5  LEFT JOIN bowling W
6  ON B.MatchID = W.MatchID AND B.PID = W.PID
7  union
8  SELECT B.MatchID, B.PID, B.NRuns, W.NOvers, W.Nwickets
9  FROM batting B
10 RIGHT JOIN bowling W
11 ON B.MatchID = W.MatchID AND B.PID = W.PID;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| MatchID | PID | NRuns | NOvers | Nwickets |
|---|---|---|---|---|
| 2755 | 23001 | 0 | 10 | 2 |
| 2755 | 25001 | 71 | NULL | NULL |
| 2755 | 91001 | 60 | 4 | 0 |
| 2755 | 94002 | 17 | 9 | 1 |
| 2755 | 92002 | 1 | 10 | 1 |
| 2689 | 89001 | 91 | NULL | NULL |
| 2689 | 23001 | 38 | 3 | 0 |
| 2689 | 25001 | 36 | NULL | NULL |
| 2689 | 99002 | 2 | NULL | NULL |
| 2689 | 95001 | 1 | NULL | NULL |
| 2689 | 24001 | 42 | 3 | 1 |
| 2689 | 99001 | 7 | 10 | 1 |

**6 [3-Table Inner Join]:** List Player.FName, Player.LName, ODIMatch.Ground, ODIMatch.Date, and Batting.NRuns for all matches played where 'India' was either Team1 or Team2.

```
SELECT P.FName, P.LName, M.Ground, M.Date, B.NRuns
FROM player P
INNER JOIN batting B ON P.PlayerID = B.PID
INNER JOIN odimatch M ON B.MatchID = M.MatchID
WHERE M.Team1 = 'India' OR M.Team2 = 'India';
```

Query 1

Limit to 1000 rows

```
1  USE anmol_11;
2
3  SELECT P.FName, P.LName, M.Ground, M.Date, B.NRuns
4  FROM player P
5  INNER JOIN batting B ON P.PlayerID = B.PID
6  INNER JOIN odimatch M ON B.MatchID = M.MatchID
7  WHERE M.Team1 = 'India' OR M.Team2 = 'India';
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| FName | LName | Ground | Date | NRuns |
|---|---|---|---|---|
| Yuvraj | Singh | Colombo | 2008-08-24 | 0 |
| MS | Dhoni | Colombo | 2008-08-24 | 71 |
| Sanath | Jayasuriya | Colombo | 2008-08-24 | 60 |
| Chaminda | Vaas | Colombo | 2008-08-24 | 17 |
| Muthiah | Murlitharan | Colombo | 2008-08-24 | 1 |
| Sachin | Tendulkar | Brisbane | 2008-03-04 | 91 |
| Yuvraj | Singh | Brisbane | 2008-03-04 | 38 |
| MS | Dhoni | Brisbane | 2008-03-04 | 36 |
| Adam | Gilchrist | Brisbane | 2008-03-04 | 2 |
| Ricky | Ponting | Brisbane | 2008-03-04 | 1 |
| Andrew | Symonds | Brisbane | 2008-03-04 | 42 |
| Breet | Lee | Brisbane | 2008-03-04 | 7 |
| Praveen | Kumar | Brisbane | 2008-03-04 | 7 |
| Praveen | Kumar | Colombo | 2008-08-24 | 2 |
| Harbhajan | Singh | Brisbane | 2008-03-04 | 3 |
| Harbhajan | Singh | Colombo | 2008-08-24 | 2 |

**7 [4-Table Multi-Join]:** Display player full name, Ground, Date, Batting.NRuns, and Bowling.Nwickets for all players who both BATTED and BOWLED in the exact same match.

```
SELECT P.FName, P.LName, M.Ground, M.Date,
    B.NRuns, W.Nwickets
FROM player P
INNER JOIN batting B ON P.PlayerID = B.PID
INNER JOIN bowling W
ON B.MatchID = W.MatchID AND B.PID = W.PID
INNER JOIN odimatch M ON B.MatchID = M.MatchID;
```

Query 1

Limit to 1000 rows

```
1  USE anmol_11;
2
3  SELECT P.FName, P.LName, M.Ground, M.Date,
4         B.NRuns, W.Nwickets
5  FROM player P
6  INNER JOIN batting B ON P.PlayerID = B.PID
7  INNER JOIN bowling W
8  ON B.MatchID = W.MatchID AND B.PID = W.PID
9  INNER JOIN odimatch M ON B.MatchID = M.MatchID;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| FName | LName | Ground | Date | NRuns | Nwickets |
|---|---|---|---|---|---|
| Breet | Lee | Brisbane | 2008-03-04 | 7 | 1 |
| Andrew | Symonds | Brisbane | 2008-03-04 | 42 | 1 |
| Yuvraj | Singh | Brisbane | 2008-03-04 | 38 | 0 |
| Chaminda | Vaas | Colombo | 2008-08-24 | 17 | 1 |
| Muthiah | Murlitharan | Colombo | 2008-08-24 | 1 | 1 |
| Sanath | Jayasuriya | Colombo | 2008-08-24 | 60 | 0 |
| Yuvraj | Singh | Colombo | 2008-08-24 | 0 | 2 |
| Harbhajan | Singh | Colombo | 2008-08-24 | 2 | 3 |
| Harbhajan | Singh | Brisbane | 2008-03-04 | 3 | 1 |

**8 [Self Join - Attribute Match]:** Write a query using a SELF JOIN on the Player table to find pairs of players who were born in the same birth place (Bplace). Display Player 1 Name, Player 2 Name, and Bplace (exclude self-pairings and duplicate reversed pairs).

SELECT P1.FName AS Player1, P2.FName AS Player2, P1.Bplace

FROM player P1

INNER JOIN player P2 ON P1.Bplace = P2.Bplace

WHERE P1.PlayerID < P2.PlayerID;

```
1  USE anmol_11;
2
3  SELECT P1.FName AS Player1, P2.FName AS Player2, P1.Bplace
4  FROM player P1
5  INNER JOIN player P2 ON P1.Bplace = P2.Bplace
6  WHERE P1.PlayerID < P2.PlayerID;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| Player1 | Player2 | Bplace |
|---|---|---|
| Virender | Ishant | Delhi |
| Gary | Jacques | Cape Town |
| Gary | Herschelle | Cape Town |
| Jacques | Herschelle | Cape Town |

**9 [Self Join - Match Analysis]:** Using a SELF JOIN on ODIMatch, find pairs of distinct matches that were played on the same Ground on different dates.

SELECT M1.MatchID AS Match1, M2.MatchID AS Match2,

M1.Ground, M1.Date AS Date1, M2.Date AS Date2

FROM odimatch M1

INNER JOIN odimatch M2 ON M1.Ground = M2.Ground

WHERE M1.Date <> M2.Date

AND M1.MatchID < M2.MatchID;

```
1 •  USE anmol_11;
2
3 •  SELECT M1.MatchID AS Match1, M2.MatchID AS Match2,
4           M1.Ground, M1.Date AS Date1, M2.Date AS Date2
5    FROM odimatch M1
6    INNER JOIN odimatch M2 ON M1.Ground = M2.Ground
7    WHERE M1.Date <> M2.Date
8    AND M1.MatchID < M2.MatchID;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| Match1 | Match2 | Ground | Date1 | Date2 |
|---|---|---|---|---|
| 2404 | 2750 | Colombo | 2006-08-16 | 2008-08-24 |
| 2404 | 2755 | Colombo | 2006-08-16 | 2008-08-24 |

**10 [Cross Join]:** Generate a Cartesian product (CROSS JOIN) between players from 'India' and all matches in ODIMatch to display all hypothetical player-match combinations.

SELECT P.FName, P.LName, M.MatchID, M.Ground, M.Date

FROM player P

CROSS JOIN odimatch M

WHERE P.Country = 'India';

```sql
USE anmol_11;

SELECT P.FName, P.LName, M.MatchID, M.Ground, M.Date
FROM player P
CROSS JOIN odimatch M
WHERE P.Country = 'India';
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| FName | LName | MatchID | Ground | Date |
|---|---|---|---|---|
| Harbhajan | Singh | 2201 | Guwahati | 2007-11-05 |
| Saurav | Ganguly | 2201 | Guwahati | 2007-11-05 |
| Rahul | Dravid | 2201 | Guwahati | 2007-11-05 |
| Anil | Kumble | 2201 | Guwahati | 2007-11-05 |
| Sachin | Tendulkar | 2201 | Guwahati | 2007-11-05 |
| Ishant | Sharma | 2201 | Guwahati | 2007-11-05 |
| Praveen | Kumar | 2201 | Guwahati | 2007-11-05 |
| MS | Dhoni | 2201 | Guwahati | 2007-11-05 |
| Yuvraj | Singh | 2201 | Guwahati | 2007-11-05 |
| Virender | Sehwag | 2201 | Guwahati | 2007-11-05 |
| Harbhajan | Singh | 2324 | Peshwar | 2006-02-06 |
| Saurav | Ganguly | 2324 | Peshwar | 2006-02-06 |
| Rahul | Dravid | 2324 | Peshwar | 2006-02-06 |
| Anil | Kumble | 2324 | Peshwar | 2006-02-06 |
| Sachin | Tendulkar | 2324 | Peshwar | 2006-02-06 |
| Ishant | Sharma | 2324 | Peshwar | 2006-02-06 |
| Praveen | Kumar | 2324 | Peshwar | 2006-02-06 |
| MS | Dhoni | 2324 | Peshwar | 2006-02-06 |
| Yuvraj | Singh | 2324 | Peshwar | 2006-02-06 |
| Virender | Sehwag | 2324 | Peshwar | 2006-02-06 |
| Harbhajan | Singh | 2327 | Rawalpindi | 2006-02-11 |

Result 16

**11 [Natural Join vs Explicit]:** Write an SQL query joining Batting and ODIMatch using MatchID to display total runs scored by each player in matches played at 'Colombo' ground.

SELECT B.PID, SUM(B.NRuns) AS Total_Runs

FROM batting B

INNER JOIN odimatch M ON B.MatchID = M.MatchID

WHERE M.Ground = 'Colombo'

GROUP BY B.PID;

```sql
USE anmol_11;

SELECT B.PID, SUM(B.NRuns) AS Total_Runs
FROM batting B
INNER JOIN odimatch M ON B.MatchID = M.MatchID
WHERE M.Ground = 'Colombo'
GROUP BY B.PID;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| PID | Total_Runs |
|---|---|
| 23001 | 0 |
| 25001 | 71 |
| 91001 | 60 |
| 94002 | 17 |
| 92002 | 1 |
| 27001 | 2 |
| 98002 | 2 |

**12 [Join with Aggregation]:** Display PlayerID, FName, LName, and total career runs (SUM(NRuns)) across all matches by joining Player and Batting, sorted in descending order of total runs.

SELECT P.PlayerID, P.FName, P.LName,

SUM(B.NRuns) AS Total_Runs

FROM player P

INNER JOIN batting B ON P.PlayerID = B.PID

GROUP BY P.PlayerID, P.FName, P.LName

ORDER BY Total_Runs DESC;

```
1 •  USE anmol_11;
2
3 •  SELECT P.PlayerID, P.FName, P.LName,
4           SUM(B.NRuns) AS Total_Runs
5    FROM player P
6    INNER JOIN batting B ON P.PlayerID = B.PID
7    GROUP BY P.PlayerID, P.FName, P.LName
8    ORDER BY Total_Runs DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| PlayerID | FName | LName | Total_Runs |
|---|---|---|---|
| 25001 | MS | Dhoni | 107 |
| 89001 | Sachin | Tendulkar | 91 |
| 91001 | Sanath | Jayasuriya | 60 |
| 24001 | Andrew | Symonds | 42 |
| 23001 | Yuvraj | Singh | 38 |
| 94002 | Chaminda | Vaas | 17 |
| 27001 | Praveen | Kumar | 9 |
| 99001 | Breet | Lee | 7 |
| 98002 | Harbhajan | Singh | 5 |
| 99002 | Adam | Gilchrist | 2 |
| 92002 | Muthiah | Murlitharan | 1 |
| 95001 | Ricky | Ponting | 1 |

**13 [Left Join Anti-Join]:** Write an SQL query using LEFT JOIN to identify all players (PlayerID, FName, LName, Country) who have NEVER recorded a bowling performance in the Bowling table.

SELECT P.PlayerID, P.FName, P.LName, P.Country

FROM player P

LEFT JOIN bowling B ON P.PlayerID = B.PID

WHERE B.PID IS NULL;

```sql
USE anmol_11;

SELECT P.PlayerID, P.FName, P.LName, P.Country
FROM player P
LEFT JOIN bowling B ON P.PlayerID = B.PID
WHERE B.PID IS NULL;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| PlayerID | FName | LName | Country |
|---|---|---|---|
| 21001 | Virender | Sehwag | India |
| 25001 | MS | Dhoni | India |
| 27001 | Praveen | Kumar | India |
| 27002 | Ishant | Sharma | India |
| 89001 | Sachin | Tendulkar | India |
| 90001 | Brain | Lara | West Indies |
| 90002 | Anil | Kumble | India |
| 92001 | Shane | Warne | Australia |
| 92003 | Inzamam | UI-Huq | Pakistan |
| 93002 | Heath | Streak | Zimbabwe |
| 93003 | Gary | Kirsten | South Africa |
| 94004 | Stephen | Fleming | New Zealand |
| 95001 | Ricky | Ponting | Australia |
| 95002 | Shaun | Pollock | South Africa |
| 95003 | Jacques | Kallis | South Africa |
| 96001 | Rahul | Dravid | India |
| 96002 | Herschelle | Gibbs | South Africa |
| 96003 | Saurav | Ganguly | India |
| 97004 | Daniel | Vettori | New Zealand |
| 98001 | Shahid | Afridi | Pakistan |
| 99002 | Adam | Gilchrist | Australia |

Result 19